

Geometric Acoustics and Diffuse Sound Fields

So far in this book we dealt with sound propagation in terms of the wave equation. This procedure becomes very complicated, however, when treating sound fields inside rooms with complicated shapes like concert halls or churches. An approximate method called *geometrical acoustics* is often useful in these cases.

This method considers sound propagation in terms of so-called *sound rays*. Sound rays were already introduced in Section 10.4, where a ray symbol was used to designate the wave bundle that emerges from a circular hole in a rigid wall when $R \gg \lambda$. The idea is that the wave bundle propagates along a straight line like a ray of light.

The concept of rays is mathematically achieved by maintaining plane areas of constant phase and letting the wavelength go to zero. In praxi, wave propagation can be approximated by rays when the following condition is met. The wavelength of the sound under consideration must be small compared to the linear dimensions of boundary areas and obstacles. Diffraction is neglected in this view.

The energy density, W'' , within a ray is equal to the energy density in a plane propagating wave. To compute its amount, we consider a wave bundle propagating through an area of $1/\text{m}^2$ for 1 m – see Fig. 12.1 for illustration. The energy density, then, is the active power, $\bar{P}$, times the traveling time, $t_1 = (1/c) \cdot 1\text{ m}$, divided by the volume of 1 m^3 , or – in mathematical terms,

$$W''_{\text{ray}} = \underbrace{\underbrace{|\vec{I}| \cdot 1\text{ m}^2}_{\bar{P}} \underbrace{\frac{1}{c} \cdot 1\text{ m}}_{t_1}}_W \underbrace{\frac{1}{1\text{ m}^3}}_{V^{-1}} = \frac{|\vec{I}|}{c} = \frac{\bar{I}}{c}. \quad (12.1)$$

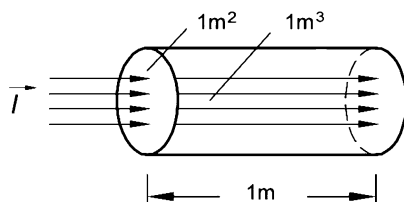


Fig. 12.1. Sound rays propagating through a unit volume

Rays are usually considered to be incoherent so that their energy densities superimpose when they meet¹. The sum up of the rays is

$$\sum W''_{\text{ray}} = \frac{\sum |\vec{I}|}{c} = \frac{\bar{I}_{\Sigma}}{c}. \quad (12.2)$$

The assumption that the rays are incoherent is valid for most broadband signals like speech or music, assuming that the rays have traveled different distances from the source. This is not the case, however, for impinging and reflected waves close to reflecting surfaces. Incoherence can also not be assumed for narrow-band or pure-tone signals.

12.1 Mirror Sound Sources and Ray Tracing

The behavior of rays at plane reflecting surfaces is particularly relevant for geometrical acoustics. Plane means here, that any unevenness of the surface is small compared to the wavelengths of the sound considered. The reflection law, $\theta_1 = \theta'_1$ holds, and may even be applied to slightly curved planes as long as the curvature is small compared to the wavelength – shown in Fig. 12.2.

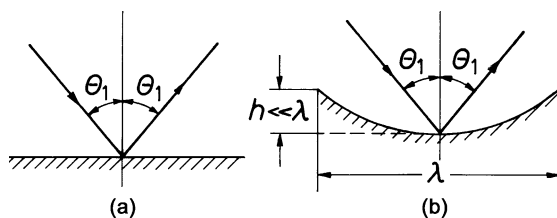


Fig. 12.2. Reflection of sound rays at (a) planes, and (b) moderately curved plates

It is particularly useful to treat sound propagation by using rays when studying room acoustics and outdoor sound propagation over longer distances – as often necessary in connection with noise control problems. When using the

¹ Consult Section 1.6 for incoherent superposition

concept of sound rays, relevant rules and laws from optics can directly be applied. The length of a ray is proportional to its traveling time, making it possible to not only determine the direction of sound propagation, but also the arrival times of different rays at a certain point of interest.

Reflection on plane surfaces can be depicted by *mirror sources* (*virtual sources*) – shown in Fig. 12.3. The mirror source, $\underline{q}_m$, and the primary source, $\underline{q}_0$, simultaneously send out identical sound fields. The combination of these sound fields on the surface produces a reflected wave that fulfills the boundary condition for full reflection, namely, the normal component of the particle velocity, $v_\perp$, being zero.

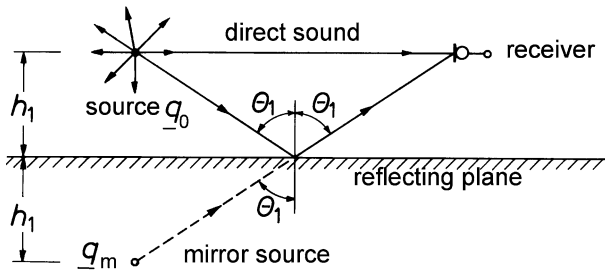


Fig. 12.3. Mirror sound sources emanating from reflection at a plane

In Figure 12.4, it is assumed that both sources transmit a short sound impulse at the initial time, t_0 . The figure shows the wave fronts of both the primary and the reflected sounds and illustrates how the second wave front arrives at the receiver later than the first.

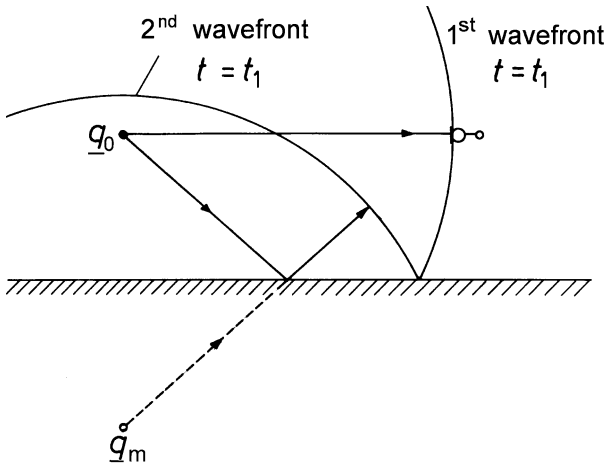


Fig. 12.4. Wave fronts of both the primary and the reflected sounds

Investigations into the relative arrival times of reflections are important, especially since reflections that arrive at the receiver with a delay may cause the perception of disturbing echoes, which should be avoided in room acoustics.

The actual perceptual *echo threshold* is dependent on the character of the sound. It is about 50 ms for running speech, larger for music and shorter for impulses.

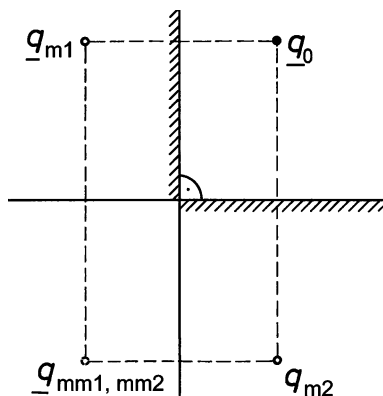


Fig. 12.5. Mirror sources at edges and in corners

Complications may arise for higher order reflections on the edges and in the corners of rectangular spaces. Mirror sources may coincide spatially. This is illustrated in Fig. 12.5 where the 2nd-order mirror sources, $\underline{q}_{mm1}$ and $\underline{q}_{mm2}$, coincide in the rectangular corner.

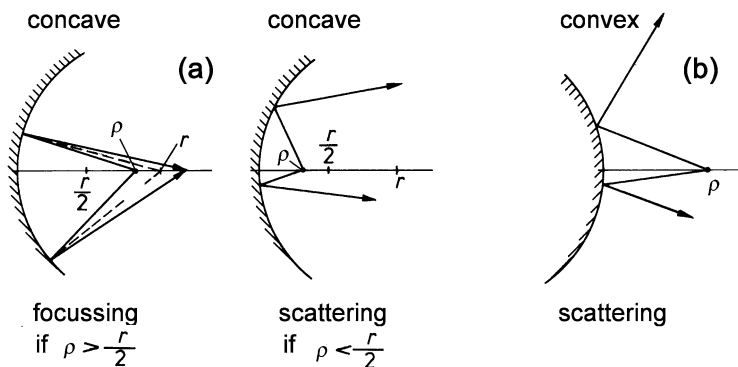


Fig. 12.6. Focussing or scattering effects (ρ denotes the source position) (a) at concave, (b) at convex surfaces

Focussing or scattering may happen as a result of curved surfaces, – shown in Fig. 12.6. Unwanted echoes caused by focussing can be avoided by modifying the form of the reflecting surface, employing irregular reflecting structures with linear dimensions that are on the order of the wavelength, or by covering the surfaces with sound-absorbing materials.

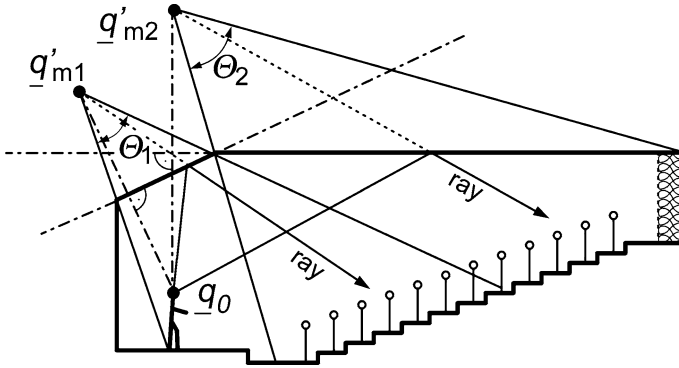


Fig. 12.7. Guiding sound via the ceiling of an auditorium

Figure 12.7 presents an application example for geometrical acoustics. In an auditorium, the sound from a speaker, q_0 , is guided to the audience via ceiling reflections. q'_{m1} and q'_{m2} are the mirror sound sources representing the tilted and horizontal parts of the ceiling, respectively. Two sample rays are depicted for illustration. The two mirror sources illuminate the spatial sections Θ_1 and Θ_2 . The rear wall is made absorptive to avoid audible echoes.

As to the construction of the graph please note that the mirror sources are positioned perpendicularly to the reflecting surfaces at the same distance to the surface as the original source, yet, outside the room under consideration. The rays originating from them are restricted to the spatial sector defined by the individual reflecting surfaces concerned.

12.2 Flutter Echoes

We will now consider a case involving a highly directional sound source, p_1 , between two parallel walls a distance, l , apart. The source emits a short sound-pressure impulse directed perpendicularly toward one of the walls and propagating like a ray – shown in Fig. 12.8(a). The two walls may be slightly absorbent, characterized by a degree of absorption, α . A microphone close to the position of the source would record a signal as schematically plotted in Fig. 12.8(b).

If the interval between the individual impulses at the receiver, $\tau = l/c$, is larger than the echo threshold, the impulses become perceptible as a series

of individual echoes, called *flutter echo*. Flutter echoes should be avoided in room acoustics. This can be accomplished by slightly tilting the two walls by $> 5^\circ$ or by making their surfaces absorbing or scattering.

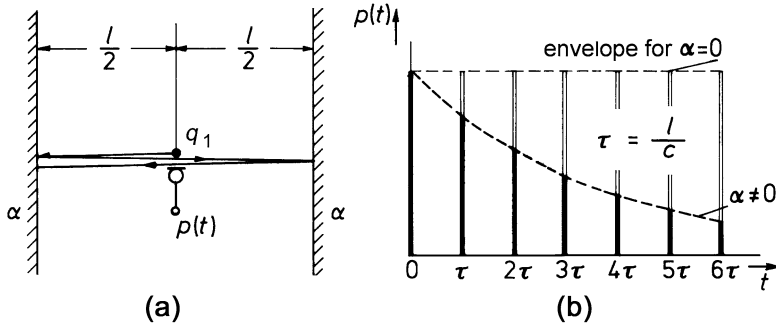


Fig. 12.8. Multiple reflections between parallel walls – the origin of flutter echoes

The envelope of the impulse series decreases exponentially for $\alpha > 0$, meaning that the ray loses a given percentage of its energy whenever a reflection takes place². The following variables are involved in the flutter echo situation, with W_0'' being the energy density of the impinging ray,

$$\begin{aligned} W_1'' &= (1 - \alpha) W_0'' \quad \dots \text{energy density after the 1st reflection} \\ W_n'' &= (1 - \alpha)^n W_0'' \quad \dots \text{energy density after the } n^{\text{th}} \text{ reflection} \end{aligned}$$

With l being the traveled distance between two reflections, we arrive at a temporal *reflection density*, n' , that is the number of reflections per time, of

$$n' \approx c/l. \quad (12.3)$$

Please note that n' is correct up to a rest that is negligible when $l \ll 340\text{m}$ holds. 340 m is the distance that sound propagates in air in one second.

For large n , the expression for W_n'' can be substituted with a monotonic function as follows,

$$W''(t) = W_0'' (1 - \alpha)^{n' t} = W_0'' (1 - \alpha)^{(c/l) t}. \quad (12.4)$$

The discrete energy losses are replaced in this way by continuous spatial damping, which allows the above expression to be written using $y = e^{\ln y}$, namely,

$$W''(t) = W_0'' \left[e^{\ln(1-\alpha)} \right]^{n' t} = W_0'' e^{n' \ln(1-\alpha) t}. \quad (12.5)$$

² The chunks of sound energy that the source sends out and that subsequently oscillate between the two walls are sometimes dubbed “sound particles”

This decreasing exponential function is actually an analytical description of the decreasing envelope in Fig. 12.8 (b).

For very small amounts of absorption, $\alpha \ll 1$, the expression can be simplified by using the serial expansion

$$-\ln(1 - \alpha) = \alpha + \frac{\alpha^2}{2} + \frac{\alpha^3}{3} + \dots \approx \alpha \quad (12.6)$$

and truncating it after the first term, so that

$$W''(t) \approx W_0'' e^{-n' \alpha t}. \quad (12.7)$$

12.3 Impulse Responses of Rectangular Rooms

We will now move beyond the case of two parallel walls and consider a rectangular (cuboid) room with six reflecting boundaries, namely, four walls, one floor and one ceiling. This room will illustrate an important rule in room acoustics that which states that the reflection density, n' , increases with t^2 in many rooms.

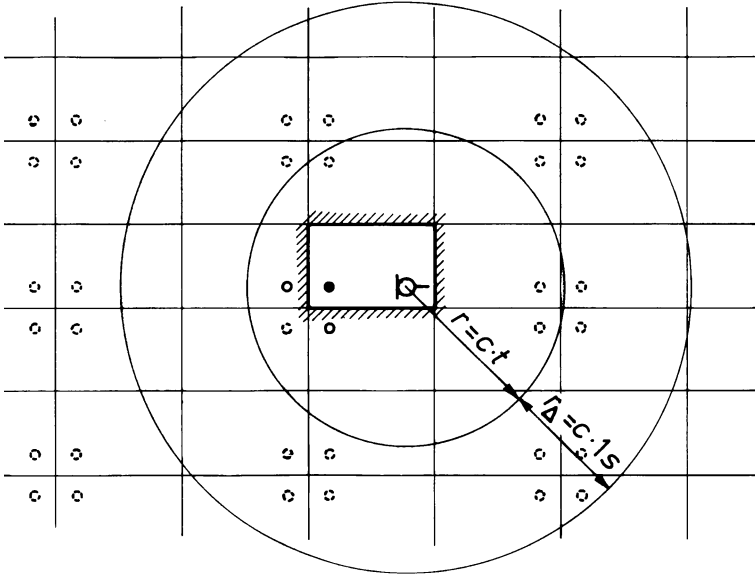


Fig. 12.9. Image sources of one sound source for a rectangular room

Figure 12.9 illustrates this concept. The figure shows the plan of a rectangular room with a single sound source in it, along with mirrored rooms of n^{th} order with one mirror source in each of them³.

Let V be the volume of the cuboid room. Now all of the mirror sources simultaneously transmit a sound impulse at $t = 0$. All impulses that originate from within a hollow sphere with the radius $r_{\Delta} = c \cdot 1 \text{ s}$, arrive at the receiver in the original room within the same interval of 1 s. The number of mirror sources in the hollow sphere is approximately the volume of the hollow sphere divided by the volume of the original cuboids, V , which is

$$n \approx \frac{\frac{4}{3} \pi [(r + r_{\Delta})^3 - r^3]}{V}. \quad (12.8)$$

For $r \ll r_{\Delta}$, neglecting higher-order difference terms, r_{Δ}^2 and r_{Δ}^3 , this expression approaches

$$n'(t) = \frac{4\pi r^2 r_{\Delta}}{V \cdot 1 \text{ s}} = \frac{4\pi c^3 t^2}{V}. \quad (12.9)$$

In other words, the density of the impulses arriving at the receiver is increasing with the square of expired time. The reflections also come from more directions over time, resulting in an ever more homogeneous distribution both over time and space.

Figure 12.10 is a simplified illustration of what is called an *echogram*, particularly, an impulse echogram. We see the direct sound and the early, low-order reflections as discrete event. Then the echogram becomes denser and denser, so that individual impulses can no longer be discriminated. This late part of the echogram is called *reverberation tail*.

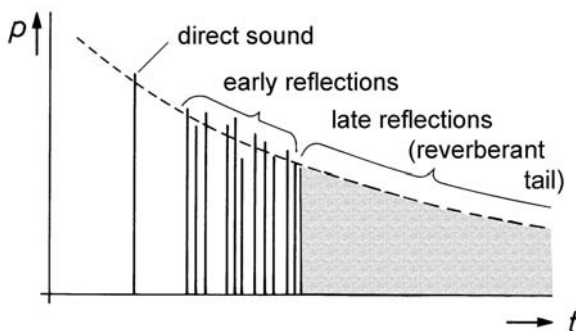


Fig. 12.10. Echogram

It is important to consider the intensity characteristics of this situation. Assuming that the sound source is a spherical source of 0th order, the active

³ This is the case for rectangular rooms where many mirror sources coincide spatially due to the rectangular corners. In more irregular rooms the situation may become more complicated, particularly, when focussing occurs

intensity of the transmitted sound will drop with $1/r^2$. On the one hand, this effect can be seen for the early reflections of the echogram, but on the other hand, the reflection density increases with t^2 . The active intensity measured over an interval, t_Δ , is therefore constant.

Now we have to keep in mind at this point that real instruments used to measure echograms always measure with a running time window, t_Δ , because they have low-pass characteristics. Thus, any echogram measurement would show a constant envelope as long as no absorption or dissipation occurred. Consequently, the energy in the running time window would stay constant over time. The latter is unrealistic in praxi, however, since there are always some losses. We will now show that the envelope of the reverberation tail decreases exponentially in real rooms.

12.4 Diffuse Sound Fields

From the discussion in Section 12.3, it is obvious that it is hardly possible to trace the fate of each individual sound ray, particularly in the reverberation tail. Nevertheless, it is possible to make important statements about the average fate of late reflections. Such an approach is called *statistical room acoustics*. We begin with an idealized model that adequately describes the sound field of the reverberant tail, also called the *diffuse sound field*. The model *diffuse sound field* is characterized by the following assumption, expressed in term of geometrical room acoustics.

A *diffuse sound field* is composed of many rays with the average properties of equal intensity and equal spatial distribution

This assumes that all rays, on average, have been reflected the same number of times and have, on average, traveled the same distance. It also means that the mean free-path length between two reflections is the same for all rays.

The results of statistical room acoustics are independent of room shape because only the average fate of rays is considered and described by statistical parameters.

Sound Power Impinging Upon the Walls

In a diffuse sound field composed of rays from all possible directions, the magnitude of the intensity is given by

$$\bar{I}_d = \iint_{4\pi} |\vec{I}(\Omega)| d\Omega = W_d'' c, \quad (12.10)$$

whereby Ω is the spatial angle and $W_d'' c$ denotes the energy density. The rays that impinge on a wall from all directions transport sound power onto the

wall. In the following we shall determine the total power that hits the walls perpendicularly.

We start by computing the active intensity that hits a small surface element, dA , on the wall. This intensity is obtained by integration over all differential intensities on a hemisphere of – as depicted in Fig. 12.11.

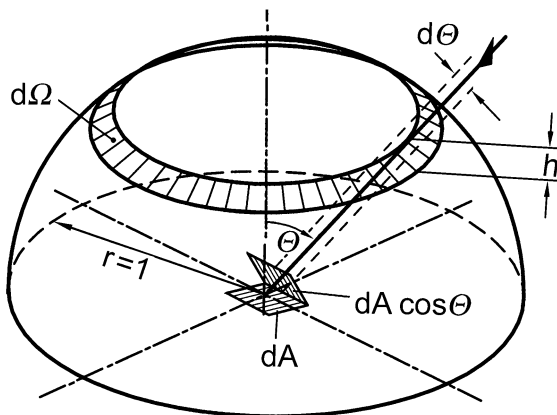


Fig. 12.11. Diffuse field impinging on a wall

The wave bundle that arrives from a spatial angle of $d\Omega$ yields the energy-density component

$$dW_d'' = \frac{W_d''}{4\pi} d\Omega, \quad (12.11)$$

where $d\Omega = 2\pi \sin \theta d\theta$ is derived from $A_{zone} = 2\pi h$, the area of a spherical zone on a unit-radius sphere. Consequently, we get

$$dI = \frac{W_d'' c}{2} \sin \theta d\theta. \quad (12.12)$$

The perpendicular component of the differential intensity, dI , is obtained as

$$dI_{wall} = dI \cos \theta. \quad (12.13)$$

To get the total intensity that hits the surface element perpendicularly, we integrate over the angle θ as follows,

$$\begin{aligned} dI_{wall} &= \frac{W_d'' c}{2} \int_0^{\pi/2} \cos \theta \sin \theta d\theta \\ &= \frac{W_d'' c}{2} \frac{1}{2} \int_0^{\pi/2} \sin 2\theta d\theta = \frac{W_d'' c}{4}. \end{aligned} \quad (12.14)$$

This is apparently only $1/4^{\text{th}}$ of the total diffuse intensity. The total power impinging on a wall with total wall area, A , thus is

$$\bar{P}_{\text{wall}} = \frac{W_{\text{d}}'' c}{4} A. \quad (12.15)$$

Sound Power Absorbed by the Walls

The power absorbed by the walls is

$$\bar{P}_{\text{wall, abs}} = \frac{W_{\text{d}}'' c}{4} \bar{\alpha} A, \quad (12.16)$$

where $\bar{\alpha}$ is the degree of absorption for diffuse sound incidence. It can be determined by the equation

$$\bar{\alpha} = \int_0^{\pi/2} \alpha(\Theta) \sin 2\Theta d\Theta. \quad (12.17)$$

This expression is known as *Paris'* formula and has already been mentioned in Section 11.4. The validity of *Paris'* formula becomes clear by realizing that the absorbed power is equal to

$$dI_{\text{wall, abs}}(\Theta) = \alpha(\Theta) dI_{\text{wall}}. \quad (12.18)$$

Average Reflection Density and Average Free-path Length

The energy that hits the walls during a time span of one second is

$$W_{1\text{s}} = \frac{W_{\text{d}}'' c}{4} A \cdot 1\text{s}. \quad (12.19)$$

The total diffuse-field energy present in a room with the volume, V , is

$$W_{\text{room}} = W_{\text{d}}'' V. \quad (12.20)$$

The rays transport this energy to the wall on an average of $\bar{n}'$ times per second, that is

$$\bar{n}' W_{\text{d}}'' V = \frac{W_{\text{d}}'' c}{4} A \cdot 1\text{s}, \quad \text{with } \bar{n}' = \frac{A c}{4V}. \quad (12.21)$$

This leads to the expression for the mean free-path length as

$$\bar{l} = \frac{c}{\bar{n}'} = \frac{4V}{A}. \quad (12.22)$$

12.5 Reverberation-Time Formulae

When we insert the average reflection density $\bar{n}'$, and the average degree of absorption, $\bar{\alpha}$, into (12.5), which describes the decay of a multiply reflected wave bundle – see Section 12.2 – we obtain the following expression for the time-dependency of the energy density in the diffuse sound field

$$W_d''(t) = \sum W_{\text{ray}}''(t) = W_d''(t=0) \exp \left[\frac{Ac}{4V} \ln(1 - \bar{\alpha}) t \right]. \quad (12.23)$$

According to *Sabine*, the time span required for the energy density to decrease to 1 millionth of its initial value, that is, by 10^6 or 60 dB, is called the *reverberation time*, T . Inserting this reverberation time into (12.23) results in

$$10^{-6} W_d''(t=0) = W_d''(t=0) \exp \left[\frac{Ac}{4V} \ln(1 - \bar{\alpha}) T \right]. \quad (12.24)$$

In air under normal condition, that is 20° C temperature and 1000 hPa static pressure, the sound speed, c , is about 340 m/s. Using this value, we obtain

$$T = 0.163 \left[\frac{\text{s}}{\text{m}} \right] \frac{V}{-\ln(1 - \bar{\alpha}) A}, \quad (12.25)$$

an expression that is known as *Eyring's* reverberation formula. For small amounts of absorption, $\bar{\alpha} \ll 1$, this formula simplifies to

$$T \approx 0.163 \left[\frac{\text{s}}{\text{m}} \right] \frac{V}{\bar{\alpha} A}, \quad (12.26)$$

which is preferred in praxi and known as *Sabine's* reverberation formula.

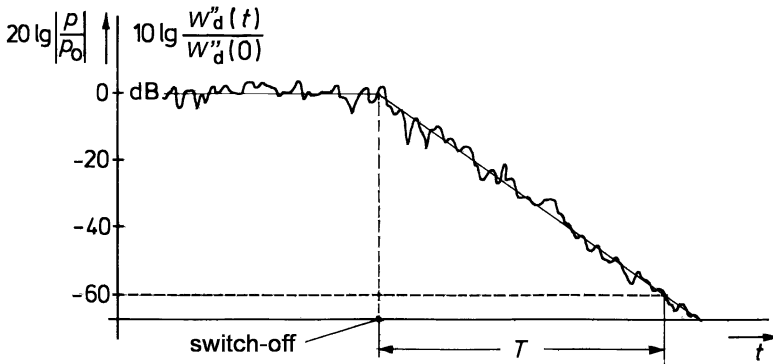


Fig. 12.12. Time trace of a reverberant noise sound after being switched off

The reverberation time, T , can be measured in many ways, including the following *switch-off method*. A room is excited with a noise source until a

stationary average sound-pressure level is reached. Then the sound source is switched off, and the sound-pressure level is recorded as a function of time, producing a curve like the one shown in Fig. 12.12. The time interval between switching-off the source and the instance where the curve has decreased by 60 dB, is taken to be the reverberation time, T .

Consideration of the theory an actual reverberation plots shows that the *diffuse sound field* assumption is sufficiently valid when the following conditions are met.

- Sound absorption is well distributed about the boundaries of the room
- The shape of the room is irregular and focussing elements are particularly avoided
- Total sound absorption is small or moderate. A good check is if the ratio of room volume and equivalent absorptive area – see next paragraph – is not too small, usually > 1 m

The Equivalent Absorptive Area

The following expression,

$$A_\alpha = -\ln(1 - \bar{\alpha}) A \approx \bar{\alpha} A, \quad (12.27)$$

is understood to be a fictive area with a unit degree of absorption, that is 100% of absorption or $\bar{\alpha} = 1$, representing the total absorption present in the room. This fictive area is called *equivalent absorptive area*.

If there are partial areas at the boundary of a room, each with an individual absorption value of $\bar{\alpha}_i$, then the equivalent absorptive area is defined to be

$$A_\alpha = \sum_i A_i \bar{\alpha}_i + \underbrace{8 \bar{\alpha} V}_{\text{correction}}, \quad (12.28)$$

where A_i are the individual areas with individual degrees of absorption $\bar{\alpha}_i$.

The correction component in the formula, not derived here, is only used to account for dissipation in the transmission medium of large rooms. Values for air are given in Table 11.1.

12.6 Application of Diffuse Sound Fields

We will now introduce three well known and frequently used applications of the diffuse-sound-field model.

Reverberation-time Acoustics

The reverberation time, T , estimated with either *Eyring's* or *Sabine's* formula, is considered to be a relevant parameter of the *acoustic quality* of spaces, which includes, among other things, their suitability for specific performances.

Table 12.1 presents preferred values of T for various performance styles. The values are drawn from literature and refer to the 500–1000 Hz range. A moderate increase toward low frequencies is considered adequate since it is said to increase the listeners' sense of envelopment and warmth.

Table 12.1. Reverberation times

Speech	Chamber music	Opera houses	Concert halls	Organ music
0.8–1.0 s	1.4–1.6 s	1.5–1.7 s	1.9–2.2 s	2.5 s and more

The following guidance holds for speech. A too-short reverberation time produces high intelligibility but also increases the effort required from the speaker. If T is too long, auditory smearing takes place and deteriorates intelligibility. 0.8–1.0 s is a reasonable compromise for speech running at normal speed, which is roughly 50 syllables/minute.

Reverberation time is undoubtedly an important parameter of acoustic quality but certainly not the only one. Proper guidance of early reflections and avoidance of echoes are at least as important.

Measurement of Spatially Averaged Absorption

Absorption measurements are taken in *reverberation chambers*, which are rooms where a diffuse sound field has been realized using highly reflective, obliquely oriented walls and planes⁴.

The equivalent absorptive area, A_{α_1} , of the empty chamber must be determined beforehand, usually by a reverberation-time measurement. With a sample of the material to be measured in the chamber, the reverberation-time measurement is then repeated. For the measured reverberation-time data, then, the equivalent absorptive area of the chamber including the sample is,

$$A_{\alpha_2} = 0.163 \left[\frac{\text{s}}{\text{m}} \right] \frac{V}{T}. \quad (12.29)$$

Consequently, the equivalent absorptive area of the sample results as

$$A_{\alpha, \text{sample}} = A_{\text{sample}} \bar{\alpha}_{\text{sample}} = A_{\alpha_2} - A_{\alpha_1}. \quad (12.30)$$

⁴ In order to measure α for perpendicular sound incidence only, a measuring tube may be applied – see Section 7.6

Measurement of the Total Power of a Sound Source

Sound power measurements are also undertaken in the diffuse sound field of a reverberation chamber. The source is brought into the chamber and operated to transmit ongoing sound.

The sound power transmitted by the source and the sound power absorbed by the equivalent absorptive area of the reverberation chamber will reach a stationary balance as follows⁵,

$$\underbrace{\bar{P}_{\text{source}}}_{\text{introduced}} = \underbrace{\frac{A_{\alpha} W''_{\text{d}} c}{4}}_{\text{absorbed}} \quad \text{with} \quad W''_{\text{d}} = \frac{I_{\text{d}}}{c} = \frac{p_{\text{rms}}^2}{\rho c^2}. \quad (12.31)$$

The power of the sound source, consequently, is

$$\bar{P}_{\text{source}} = p_{\text{rms}}^2 \frac{A_{\alpha}}{4 \rho c}. \quad (12.32)$$

The following rules are useful when performing the measurements. A sufficient number of measuring points must be well-distributed across the room but not too close to walls or corners because full incoherence of incoming and reflected sounds can not be guaranteed in such locations. This may result in measured values that are too high.

The Critical Radius

A stationary, diffuse sound field has the same average energy density, W''_{d} , everywhere in space. Close to a sound source, however, the energy density of the direct sound may be much higher. The situation is depicted in Fig. 12.13.

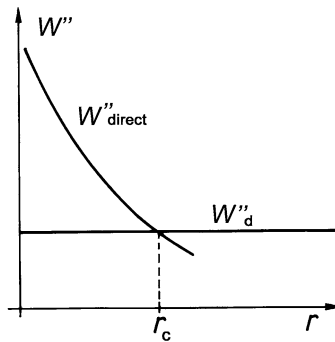


Fig. 12.13. The critical radius – equilibrium of direct- and diffuse-field energy densities with a 0th-order spherical source

⁵ With p_{rms} being the rms-value of p – see Section 15.4 for a definition

The critical radius is the distance from a 0th-order spherical source where the direct and diffuse energy densities are just equal. Equality of the two energy densities is given at

$$W''_{\text{direct}} = \underbrace{\frac{\overline{P}_{\text{source}}}{4\pi r^2}}_{I_{\text{direct}}} \frac{1}{c} \stackrel{!}{=} W''_{\text{d}} = \underbrace{\frac{4\overline{P}_{\text{source}}}{A_{\alpha}}}_{I_{\text{diffuse}}} \frac{1}{c}, \quad (12.33)$$

which leads to the critical radius as follows,

$$r_c = \sqrt{\frac{A_{\alpha}}{V}}. \quad (12.34)$$

For directional sources the distance of equilibrium of direct and diffuse field is higher in the direction of focussed transmission. We then speak of *critical distance* or *diffuse-field distance*.

Knowing the critical radius or critical distance is useful when taking sound recordings because microphone placement within this radius or distance will predominantly render direct-sound signals. Placement outside will predominantly render diffuse-sound signals, which are auditorily perceived as *spatial impression*.